

COORDINATION MEMO

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	EFA (HORIZON)	BA	UVIC	0006	X

FROM: BA TO: UVic

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SUBJECT: T507 – 7%/16.5% Simulation Model Delivery and user manual	REPLY REQUIRED: YES ☐ NO ☒ IF YES, DATE:
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1. References
2. Discussion
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1. References

None.

2. Discussion

2.1 INTRODUCTION

This memo, along with the attached data and files represents Bombardier's delivery of the T507 aerodynamic models (16.5% and 7%) to the University of Victoria Center for Aerospace Research. This delivery is part of a Horizon research project. It is intended to be used exclusively for the current project.

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This document describes the intended use of this simulation model and explains the interfaces to be used. The simulation model consists of a top level Simulink model, a Flight Simulation DLL and supporting files. Attached to this document are the files needed to run the model and an ICD, describing the interface details.

2.2 REQUIREMENTS

The top-level Simulink file has been developed in Matlab 2018b. Small modifications may be required to port the model to a newer version of Matlab. The Flight Simulation DLL is a Windows 64 bit DLL. The Model has been tested in Matlab 2018b and meets the faster than real-time requirement.

2.3 MODEL DESCRIPTION

Below is the top-level view of the model. The model consists of three parts: a bus of effector inputs, the DLLSimulation block and a bus of signal outputs. The model employs callback (PreLoadFcn, InitFcn, StartFcn, StopFcn, CloseFcn) that ensure the model is updated correctly. The configuration is set to fixed step size with a discrete solver. Changing the step size of the simulations should be done through the attached DLLSimInit.m file (see section 2.4).

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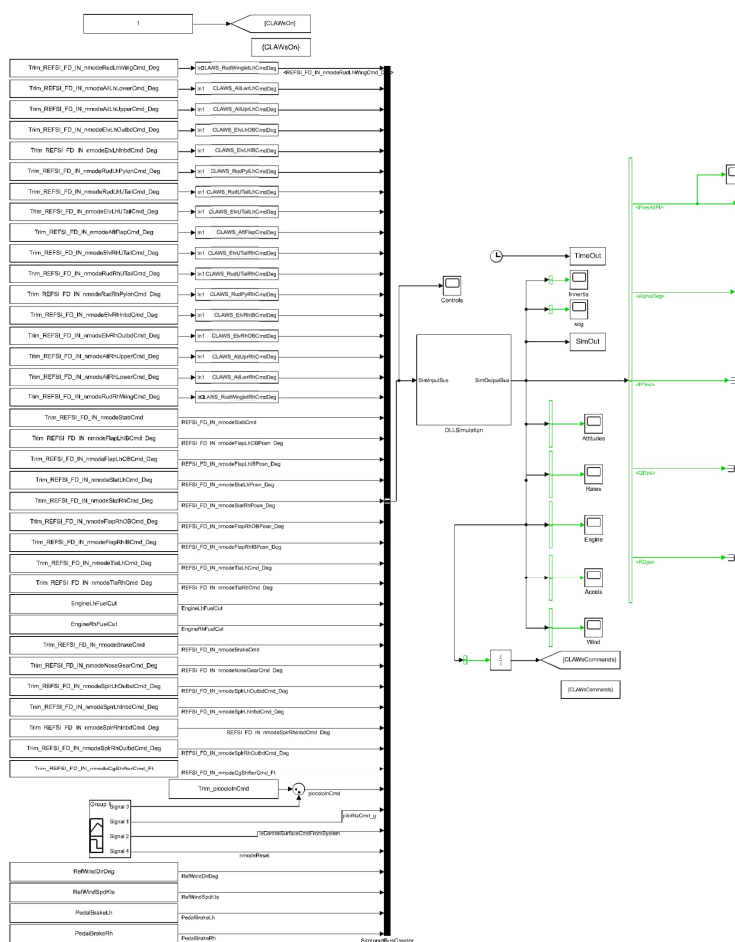


Figure 1: Top Level Model

The effector bus contains all the effector and system commands as outputted from the Control Laws. These commands are passed directly to the DLL. Inside the DLL, they will pass through an actuator model before being used by the flight dynamics. The names of the effector bus are specific and must not be changed arbitrarily. The names match parameters in the DLL that are being used as inputs and are detailed in attached Simulink ICD.

This block will convert the input bus into a vector and uses that vector to drive the simulation. Similarly, the saved signals are converted from a vector onto a bus and outputted to be used downstream. This is a wrapper for the DLL, and should not be changed. Note that initialization,

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and setup of the DLL are done in the DLLSimInit.m file (see section 2.4). No interaction is needed for this block.

2.3.3 Output Bus

The output of the DLL simulation block is a bus that contains all the sensor, flight dynamics and data signals. The outputs are defined in the DLLSimInit.m file (see section 2.4) and are detailed in attached Simulink ICD. To add or remove signals from this bus, simply change the SaveList definition in DLLSimInit.m

2.4 SIMULATION INITIALIZATION

The initialization of the model is done through DLLSimInit.m. This function is called each time the model is opened or simulated. The DLL is set to the defined flight condition and is trimmed for straight and level flight. There are four sections in this file. Only the first one should be modified. The other sections are here for transparency and to simplify changes in future revisions (all changes should be limited to this file).

2.4.1 Define Trim Setup

The first section defines the location of the DLL as well as the initial conditions of the aircraft. It is important to note that the simulation works with imperial units only. The initial conditions are implemented as Metric unit * Imperial Conversion Factor.

Note that switching between 7% and 16.5% shall only be done before the first run of a fresh Matlab session. Switching between models can be accomplished by clearing the global SimObj object, but restarting Matlab is preferable.

Comments are provided in the code itself for clarity.

Table 1: Sim Init File – Operating Point Definition

```
%% Define Trim Setup
% Beginning of User Defined Code.
% Trim Condition Definition

FTV = 3; % Set to FTV3 for 7% and FTV4 for 16.5%
if FTV==3
    Speed = 65;
elseif FTV==4
    Speed = 110;
else
    error('undefined FTV');
end

Flap = 0;% 0,1,2,3
```

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```

HStab          = 0;% Deg
Alt            = 500;% Ft
WithTail       = 1; % Set to 0 to remove Tail
xCgLocMac      =
60.77;                                     % PERCENT MAC
Track          = 10; % Deg Needs to be implemented
WindDir        = 90; % Deg
WindSpd        = 0; % knots
VertWindSpd    = 0; % knots
Longitude      = -71.624729; % Alma HWIL test Box (not to be actually
flow!)
Latitude       = 48.496366; % Degrees
Turbulence     = 0.0; % 0 = None, 1 = Light
TerrainHeight  = 100;
CGShifterInitPosn_ft = 1; % ft - initial position of CG shifter
CLAWs_On       = 1;

NWCasting     = 0; % NWCasting sets the nose wheel into casting mode

Cbar           = 0.666;
XCGMeters      = (xCgLocMac/100*Cbar + 0.37569); % cg
location relative to nose [m] WARNING: 0.37569 NEEDS TO BE UPDATED FOR 16.5%

XCG            = (XCGMeters/0.07)/0.0254 * 0.6412 - 253.52;%CG % = 0.6412*In -
253.52
YCG            = 0; % Only use 0. Alternate trim setup needed for lateral
assymetry.

if FTV==3
    Weight      = 14.545*2.2;% Lbs
    ZCG         = 0.019228/0.0254;% In inches
    IXX         = 1.417 * 3417.17;% In-lbs
    IYY         = 1.6164 * 3417.17;% In-lbs
    IZZ         = 3.0881 * 3417.17;% In-lbs
    IXZ         = 0.039009* 3417.17;% In-lbs
elseif FTV==4
    Weight      = 182.0 * 2.2;% Lbs
    ZCG         = 0.4572/0.0254;% In inches
    IXX         = 43.018 * 3417.17;% In-lbs
    IYY         = 107.98 * 3417.17;% In-lbs
    IZZ         = 144.56 * 3417.17;% In-lbs
    IXZ         = 5.8528 * 3417.17;% In-lbs
end

% 16.5% Latency
ActuatorHBL6625ResponseDelay = 0.0132;% Latency in seconds imposed on
control actuator Default shipped value 0.0132
ActuatorCBS20ResponseDelay   = 0.018; % Latency in seconds imposed on
control actuator Default shipped value 0.0180

```

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```

    ActuatorCBS15ResponseDelay = 0.018; % Latency in seconds imposed on
control actuator Default shipped value 0.0180
    ActuatorHitecSG33BLResponseDelay = 0.02; % Latency in seconds imposed on
control actuator Default shipped value 0.0200

    % 7% Latency
    SurfaceLatency = 0.000; % Latency in seconds imposed on control surfaces
Default shipped value 0.01

% In Flight trimsetup should use Piccolo. On Ground is available for
% ground operations
    PropulsionLatency = 0.050;
    BrakeLatency      = 0.200;

    EngineLhFuelCut = 0;
    EngineRhFuelCut = 0;
    RefWindDirDeg   = WindDir;
    RefWindSpdKts   = WindSpd;
    PedalBrakeLh    = 0;
    PedalBrakeRh    = 0;

    TrimSetup = 'Piccolo';
%     TrimSetup = 'OpenLoopStraightAndLevel';
%     TrimSetup = 'OpenLoopOnGround'; Speed = 0;

% Define Model
if FTV==3
    DLL_Name = '.\t507_7p_T507_7P_simt_Top3_37_2_1_Opt2.mexw64';
    ModelName = 'Sim_Runs';
elseif FTV==4
    DLL_Name = '.\t507_16p5_T507_16P5_sim_Top1_74_Opt2.mexw64';
    ModelName = 'Sim_Runs_16p5';
end

% Sets path to bin regardless of directory
addpath([pwd '\bin'])

% End of User Defined Code.

```

2.4.2 SimObj Global Declaration and Trim Setup Definitions

This section defined the DLL as an object. A global declaration is used to simplify passing the object between functions. The trim setup definition is also provided here, which for now is limited to straight and level flight (**Piccolo**) and on ground trimming (**OpenLoopOnGround**).

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It's worth noting that several parameters are defined here, and these parameters are not changed between subsequent simulations. These parameters include flap type, tail on/off, etc. If these are changed, the simulation must be restarted. As an alternative, the simulation object must as a minimum be reloaded by calling (SimObj.delete; SimObj = [];

There is a logic here to avoid re-loading the DLL with each simulation call. Loading the DLL takes about 5-10second.

Note that for the 16.5% the nose gear parameters may be changed independently of the other actuators. By default they are defined as the Hitec SG33BL actuator.

2.4.3 Trim Function Call

This section calls the trim routine and stores the post-trim results. These trim parameters are used in the simulation as inputs to the model. Currently this is limited to the effectors and a few system commands.

It should be noted that inputs to the DLL are absolute values and not relative values to trim. For example, if the input to the DLL for the horizontal stabilizer is 0, the model will receive a 0deg deflection command, and not the stab used for Trim + 0degrees. Any parameter not passed to the DLL retains its trim value.

2.4.4 Bus Definitions

This section setups the busses used by the DLL. The names in the bus must match exact parameters from the DLL or an error will occur. The output bus is extracted from the model. The signals defined in SaveList will be extracted and placed on the output bus with the appropriate name. The SaveList may be added to or removed from and will be updated in the model when the DLLSimInit is called (manually or on load or simulation).

2.5 REVISIONS

The Flight Simulation DLLs provided are equivalent to the Bombardier models T507_7P.prj rev 3.37.2.1 and T507_16P5.prj rev 1.74 and using tool version 1.54 and compiled using Matlab 2018b. The Simulink model project is revision 1.15.

This revision is intended to be used for 7% and 16.5% FTV HIL testing in preparation, respectively for FTC10 and FTC7.

It contains corrections/improvements compared to previous release:

- 7%
- Set the hinge flaps by default in the aerodynamic model. Modified flaps down/slat 0 data by advancing the maximum angle of attack. Curves are modified based on the differences predicted for slat 0 slat up vs down.

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16.5%

- Update weight shifting mechanism based on latest UVic data
 - o The WSM_cmd and NRCAActive Flags are added and accessible by UVic using REFSI_FD_IN_nmodeCgShifterCmd_Flag and REFSI_FD_IN_nmodeCgShifterNRCAActive

Simulink

- Due to the new inputs for the 16.5%, a common .slx cannot be used for both models for this release. Sim_Runs_16p5.slx is added to the project to be used with the 16.5% dll. Sim_Runs.slx is still to be used with the 7%. Proper initialization is still done within DLLSimInit.m, which is still common for both models.

3. Actions

None.

4. Attachments**4.1 SimulinkFlightSimulation_ICD_SimulationTool_RevX.xlsx**

This excel file captures the inputs and outputs of the Simulation Model.

4.2 BWB_Model_SimulationDLL_Ver38.zip

This zip file contains the Simulink model, associated DLL and supporting Files.